Task control, device binding and physical evidence

Lite3 + Z1 validated task · high-level navigation and manipulation remain scene-specific

TASK SUPERVISOR

Measured state transitions

Stand → Navigate → Settle → Reach → Press → Withdraw → Retract → Success

(a) Task controllers

Known-map navigation
Measured base pose → ONNX gait

RGB-D target localization
Unique image target → 3D surface

FK / IK + collision checks
Bounded joint path + press control

(b) Device + articulation

Joint order · units · tensor shape

RobotInterface
describe · observe · command
Joint position / effort targets

19-DOF combined articulation

Lite3 · 12 Z1 · 6 Spring · 1

(c) Physics + evidence

PhysX robot + scene
Contacts · displacement · state

Frozen traces + RGB-D + replay

Audit
20 task checks

RTX replay
Audit pass only

Measured articulation state + contact feedback

Physical port: joint_drive only; no task-time root-pose writes.

Feedback uses native runtime buffers; port.observe exposes the same state contract.

G2 root_pose is a separate kinematic adapter.

Task audit does not certify all scene assets.